

$\mathbb{Z}_q$, Groups, and the Arithmetic of Scaling

Lecture 1 of 6 — Foundations

SEC 595 — Encrypted Computing

- 1 Integer Arithmetic
- 2 Modular Arithmetic and the Ring $\mathbb{Z}_n$
- 3 Representatives, Size, and Rounding
- 4 Fixed-Point Arithmetic
- 5 Groups, Rings, Fields
- 6 The Chinese Remainder Theorem
- 7 Hardness Assumptions in Groups
- 8 The Vocabulary of Security Statements

Where we are going

Every scheme this term — ML-KEM, ML-DSA, BGV, BFV, CKKS, and the zero-knowledge protocols — computes in a ring of the form

$$R_q = \mathbb{Z}_q[X]/(X^n + 1).$$

This lecture builds the innermost layer, $\mathbb{Z}_q$.

What this lecture fixes

Four conventions that every later lecture assumes:

- how we choose **representatives**;
- how we measure the **size** of an element;
- how we **round**;
- how we **scale**.

None of it is difficult. All of it is load-bearing.

Symbol	Meaning
$[a]_q$	reduction of a mod q , <i>centered</i> range (Section 3)
$\ a\ _\infty$	infinity norm: largest absolute coefficient
$\lfloor x \rceil$	round to nearest, $\lfloor x \rceil = \lfloor x + 1/2 \rfloor$
$\lfloor x \rfloor$	truncation towards $-\infty$
q, t	ciphertext modulus, plaintext modulus
n	ring dimension — not the lattice rank

In Lecture 5, a rank- k module over a degree- n ring is a lattice of dimension nk . Different numbers.

$$\mathbb{Z} = \{ \dots, -2, -1, 0, 1, 2, \dots \}$$

Closed under three *binary operations* — each takes two inputs to one output:

- addition
- subtraction
- multiplication

Not closed under division. Repairing that is where we begin.

Theorem (division algorithm)

For $a \in \mathbb{Z}$ and $n \in \mathbb{Z}_{>0}$ there exist *unique* $q, r \in \mathbb{Z}$ with

$$a = q \cdot n + r, \quad 0 \leq r < n.$$

Input: a and n . **Output:** q and r .

We write $r = a \bmod n$.

Example — integer division

Find q and r for $a = 255$, $n = 11$.

The two restrictions

① $r \geq 0$

② $n > 0$

Drop either one and q, r are no longer unique.

$$255 = 23 \times 11 + 2 \quad \text{but also} \quad 255 = 24 \times 11 - 9.$$

When $r = 0$ we say n **divides** a , written

$$n \mid a, \quad \text{and then} \quad a = q \times n.$$

Fact 1

The integer 1 has exactly one divisor: itself.

Fact 2

Every integer greater than 1 has at least two divisors: 1 and itself. Those with exactly two are the *primes*.

Example — divisors

List all divisors of 32. How many are there?

Definition

$\gcd(a, b)$ is the largest integer dividing both a and b .

Listing divisors is hopeless for the numbers we will actually use.

Example — gcd by listing divisors

Compute $\gcd(12, 140)$.

Proposition (Euclid)

Fact 1. $\gcd(a, 0) = a$

Fact 2. $\gcd(a, b) = \gcd(b, a \bmod b)$

Iterating Fact 2 terminates: the second argument strictly decreases.

Example — Euclidean algorithm

Compute $\gcd(36, 10)$ using the two facts.

When

$$\gcd(a, b) = 1$$

we call a and b **relatively prime**, or **coprime**.

This condition will decide, repeatedly, whether something is invertible.

The extended Euclidean algorithm

The plain algorithm returns $\gcd(a, b)$.

The *extended* algorithm also returns integers u, v with

$$u \cdot a + v \cdot b = \gcd(a, b).$$

This is **Bézout's identity**.

If $\gcd(a, n) = 1$ then Bézout gives

$$ua + vn = 1,$$

so

$$ua \equiv 1 \pmod{n}, \quad \text{i.e.} \quad u = a^{-1} \text{ in } \mathbb{Z}_n.$$

This is the algorithm that actually produces inverses.

Example — inverting modulo an ML-KEM prime

Compute $17^{-1} \bmod 3329$ using the extended Euclidean algorithm, and verify your answer.

Forward reference

The same algorithm, run over *polynomials* instead of integers, inverts an element of R_q .

That is exactly the computation at the heart of the Li–Micciancio attack on CKKS.

The modulo operator

Of the two outputs of division, modular arithmetic keeps only the second:

$$a = q \times n + \boxed{r}$$

As a binary operation: $a \bmod n$, with n the **modulus**.

Example — the modulo operator

Compute:

- a. $27 \bmod 5$
- b. $36 \bmod 12$
- c. $-18 \bmod 14$

The definition demands $r \geq 0$.

So $-18 \bmod 14$ is **not** -4 .

$$-18 = (-2) \cdot 14 + 10 \quad \implies \quad -18 \bmod 14 = 10.$$

The set of residues $\mathbb{Z}_n$

Reduction mod n always lands between 0 and $n - 1$, so the operator carves out a finite set:

$$\mathbb{Z}_n = \{ 0, 1, \dots, n - 1 \}.$$

There is one $\mathbb{Z}$, but infinitely many $\mathbb{Z}_n$ — one per modulus.

Example — residue sets

Write out $\mathbb{Z}_5$ and describe $\mathbb{Z}_{100}$.

Write

$$a \equiv b \pmod{n} \quad \text{when} \quad n \mid (a - b).$$

Many integers reduce to the same residue:

$$2 \equiv 12 \pmod{10}, \quad 3 \equiv 8 \pmod{5}.$$

Congruence is an equivalence relation. It partitions $\mathbb{Z}$ into n classes

$$\bar{a} = \{a + kn : k \in \mathbb{Z}\}.$$

The set of classes is $\mathbb{Z}/n\mathbb{Z}$: we have taken $\mathbb{Z}$ and *declared n to be zero*.

Section 2.2 picked the representative $0, \dots, n-1$. Section 3 will pick a different one — nothing else changes.

Addition, subtraction and multiplication all carry over:

$$+, \quad -, \quad \times.$$

Division does not — that is what Section 2.5 is about.

Compute:

a. $(7 + 14) \bmod 15$

b. $(7 - 11) \bmod 13$

c. $(7 \times 11) \bmod 20$

Why the operations carry over

Proposition (well-definedness)

If $a \equiv a'$ and $b \equiv b' \pmod{n}$, then

$$a + b \equiv a' + b' \quad \text{and} \quad a \cdot b \equiv a' \cdot b' \pmod{n}.$$

We may pick *any* representative from each class.

$$(a + b) \bmod n = [(a \bmod n) + (b \bmod n)] \bmod n$$

$$(a - b) \bmod n = [(a \bmod n) - (b \bmod n)] \bmod n$$

$$(a \times b) \bmod n = [(a \bmod n) \times (b \bmod n)] \bmod n$$

and, by repeated multiplication,

$$a^k \bmod n = (a \bmod n)^k \bmod n.$$

This is why intermediate values never grow.

Example — reducing early

Compute

$$(1,723,345 + 2,244,945) \bmod 11$$

without ever adding the two seven-digit numbers.

Additive inverses

a and b are **additive inverses** in $\mathbb{Z}_n$ when

$$a + b \equiv 0 \pmod{n}, \quad \text{that is,} \quad b = n - a.$$

Every element has one.

Example — additive inverses

List all additive-inverse pairs in $\mathbb{Z}_{10}$.

Multiplicative inverses

a and b are **multiplicative inverses** when

$$a \times b \equiv 1 \pmod{n}.$$

Here *not* every element has one.

Example — multiplicative inverses

In $\mathbb{Z}_{10}$: find the inverse of 3.

Then try to find an inverse of 2.

Definition

$a \in \mathbb{Z}_n$ is a **unit** if it has a multiplicative inverse. The set of units is written $\mathbb{Z}_n^\times$.

An element $a \neq 0$ is a **zero divisor** if $ab \equiv 0$ for some $b \neq 0$.

Proposition

a is a unit in $\mathbb{Z}_n$ if and only if $\gcd(a, n) = 1$. Hence

$$\mathbb{Z}_n^\times = \{x \in \mathbb{Z}_n : \gcd(x, n) = 1\}.$$

Proof sketch. If $\gcd(a, n) = 1$, Bézout gives $ua + vn = 1$, so $u = a^{-1}$.
Conversely if $ab = 1 + kn$ then any common divisor of a and n divides 1. $\square$

So Section 1.4 was not a detour.

Example — the unit group

Determine $\mathbb{Z}_{10}^\times$.

Which elements are zero divisors?

When is $\mathbb{Z}_q$ a field?

Proposition

$\mathbb{Z}_q$ is a field — every nonzero element is a unit — if and only if q is prime. We then write $\mathbb{F}_q$.

But fields are not the goal

The rings we care about most this term are deliberately *not* fields:

$$R_q = \mathbb{Z}_q[X]/(X^n + 1), \quad \mathbb{Z}_Q \text{ for composite } Q.$$

They have zero divisors and non-invertible elements.

That is a feature we will exploit, not a defect we work around.

From here on we use the **centered** representative:

$$[a]_q \in \left\{ -\left\lfloor \frac{q}{2} \right\rfloor, \dots, \left\lfloor \frac{q-1}{2} \right\rfloor \right\}.$$

Same classes, different choice of representative.

Example — centered representatives

Let $q = 3329$.

What is the centered range? Compute $[3000]_{3329}$.

Why bother?

We are about to start calling elements **small**.

Out of 3329, the value 3000 is small only in the centered picture:

$$[3000]_{3329} = -329.$$

Every noise bound in every scheme this term is a statement about centered representatives.

Definition

For $a \in \mathbb{Z}_q$, $\|a\|_\infty = |[a]_q|$.

For a vector or polynomial, $\|\cdot\|_\infty$ is the largest such value over all entries or coefficients.

Example — the infinity norm

Let $q = 3329$ and take

$$a = (5, 3000, 17, 3320), \quad b = (300, 310, 295, 305)$$

in $\mathbb{Z}_{3329}^4$.

Compute $\|a\|_\infty$ and $\|b\|_\infty$. Which one is smaller?

Now compute the two Euclidean norms. Do the two measures agree on which vector is smaller?

Proposition

In $\mathbb{Z}_q$, provided no wraparound occurs,

$$\|a + b\|_{\infty} \leq \|a\|_{\infty} + \|b\|_{\infty}, \quad \|a \cdot b\|_{\infty} \leq \|a\|_{\infty} \cdot \|b\|_{\infty}.$$

Forward reference

In R_q the multiplicative bound picks up a factor of the ring dimension:

$$\|ab\|_{\infty} \leq n \|a\|_{\infty} \|b\|_{\infty},$$

because each coefficient of the product is a sum of n terms.

That factor n drives the noise growth in BFV, and is why multiplication is expensive in a way addition is not. Proved in Lecture 3.

$\lfloor x \rfloor$ truncation towards $-\infty$

$\lfloor x \rceil = \lfloor x + \frac{1}{2} \rfloor$ round to nearest

Truncation always loses magnitude in the same direction, so it accumulates **bias**.

Rounding does not.

The scaling factor Δ

We will constantly meet

$$\Delta = \left\lfloor \frac{q}{t} \right\rfloor,$$

the ratio of ciphertext modulus to plaintext modulus.

Example — the scaling factor

Let $q = 2^{30}$ and $t = 1024$.

Compute Δ . If a message $m \in \mathbb{Z}_t$ is stored as Δm , how many bits does it occupy, and how many are left underneath?

Forward reference

That picture is the **BFV** ciphertext layout: message in the most significant bits, noise below.

BGV makes the opposite choice: it scales the *noise* by t and leaves the message in the least significant bits.

CKKS merges the two and tracks the boundary with a scale factor.

Three schemes, one diagram — all about where you put Δ .

The standard trick, older than cryptography:

$$x \mapsto \lfloor \Delta \cdot x \rfloor$$

for a fixed **scaling factor** Δ — and remember Δ separately.

The $Q_{m.f}$ format

m bits for the integer part, f bits for the fractional part, with $\Delta = 2^{-f}$.

- **Resolution** (gap between representable values): 2^{-f}
- **Dynamic range**: $[-2^m, 2^m - 2^{-f}]$

Example — the $Q_{3.4}$ format

For $Q_{3.4}$, determine Δ , the resolution, and the dynamic range.

Addition versus multiplication

Let $\hat{x} = \lfloor \Delta x \rfloor$ and $\hat{y} = \lfloor \Delta y \rfloor$.

$$\hat{x} + \hat{y} \approx \Delta(x + y) \quad \text{scale } \mathbf{unchanged}$$

$$\hat{x} \cdot \hat{y} \approx \Delta^2(xy) \quad \text{scale } \mathbf{squared}$$

Multiplication leaves the result at the wrong scale.

To restore it we **rescale**: divide by Δ and round.

$$\hat{x} \cdot \hat{y} \mapsto \left\lfloor \frac{\hat{x}\hat{y}}{\Delta} \right\rfloor \approx \Delta(xy)$$

Remember this operation. It comes back as a ciphertext operation.

Example — fixed-point multiplication

Work in $Q_{4.4}$, so $\Delta = 16$.

Encode $x = 2.75$ and $y = 3.25$, multiply, and rescale. Check against the exact product.

The point of a large Δ : a small additive error is harmless.

$$\Delta\mu + e \xrightarrow{\div \Delta} \mu + \frac{e}{\Delta}$$

The error is pushed into digits we did not care about.

Example — absorbing error

Let $\mu = \pi$, $\Delta = 10^6$, and let $e = 20$ be an error.

Compute $\Delta\mu + e$, then divide back by Δ . How many correct decimal places survive?

Forward reference

CKKS takes the noise a lattice-based cryptosystem is *obliged* to add for security, and reinterprets it as the rounding error a fixed-point system was going to incur anyway.

Its `RESCALE` operation is the rescaling of this section, performed on ciphertexts.

Precision is roughly $\log \Delta - \log \|\text{noise}\|_\infty$ bits.

Definition

A **group** $\langle G, \cdot \rangle$ is a set with a binary operation satisfying:

- i) **Closure:** $a, b \in G \Rightarrow a \cdot b \in G$
- ii) **Associativity:** $(a \cdot b) \cdot c = a \cdot (b \cdot c)$
- iii) **Identity:** $\exists e$ with $a \cdot e = e \cdot a = a$
- iv) **Inverse:** $\forall a \exists \bar{a}$ with $a \cdot \bar{a} = \bar{a} \cdot a = e$

A group is **abelian** (commutative) if additionally

$$a \cdot b = b \cdot a \quad \forall a, b \in G.$$

Every group in this course is abelian.

Finite groups and order

A group is **finite** when it has finitely many elements.

Its **order** $|G|$ is that number.

Every group in this course is finite.

Definition

A **ring** $\langle R, +, \times \rangle$ is an abelian group under $+$, with a second associative operation $\times$ that distributes over $+$:

$$a \times (b + c) = (a \times b) + (a \times c).$$

It is a **commutative ring** if $\times$ commutes.

It is a **field** if every nonzero element is invertible under $\times$.

Example — naming what we already have

Which of these are groups, rings, fields? What are their orders?

$$\langle \mathbb{Z}_n, + \rangle, \quad \langle \mathbb{Z}_n^\times, \times \rangle, \quad \langle \mathbb{Z}_n, \times \rangle,$$

$$\langle \mathbb{Z}_n, +, \times \rangle, \quad \mathbb{F}_p$$

Sections 1 and 2 already established every one of these facts for $\mathbb{Z}_n$.

What the vocabulary buys is the ability to say them **once** and have them hold for

- R_q ,
- $\mathbb{Z}_n^\times$,
- the elliptic-curve groups behind the zero-knowledge material.

Definition

The **order** of $a \in G$, written $\text{ord}(a)$, is the smallest $k > 0$ with $a^k = e$.

G is **cyclic** if $G = \langle g \rangle = \{e, g, g^2, \dots, g^{|G|-1}\}$ for some g , called a **generator** — or, in $\mathbb{F}_p^\times$, a **primitive root**.

Theorem

$\mathbb{F}_p^\times$ is cyclic, of order $p - 1$.

So there is always a primitive root — an element whose powers run through every nonzero residue.

Example — a primitive root

Take $p = 13$ and $g = 2$.

List the powers of 2 modulo 13. Is 2 a primitive root? What is $\text{ord}(2)$?

Which orders occur?

Proposition

If G is cyclic of order N with generator g , then $g^{N/d}$ has order d for every $d \mid N$.

In particular, G contains an element of order d **exactly when** $d \mid N$.

An innocuous statement. It decides the shape of the NTT.

Forward reference

Fast multiplication in $\mathbb{Z}_q[X]/(X^n + 1)$ needs a primitive $2n$ -th root of unity in $\mathbb{F}_q$, which exists precisely when $2n \mid q - 1$.

$$\text{ML-KEM: } q = 3329, \quad q - 1 = 2^8 \cdot 13$$

$$\text{ML-DSA: } q = 8380417, \quad q - 1 = 2^{13} \cdot 1023$$

With $n = 256$ in both cases: ML-DSA splits into 256 linear factors; ML-KEM stops at 128 quadratic ones. That is what *incomplete NTT* means.

Computing $a^x \bmod n$ by repeated multiplication takes x steps.

Writing x in binary as $(x_{n_b-1} \cdots x_1 x_0)_2$ reduces this to $O(\log x)$:

```
SQUAREANDMULTIPLY( $a, x, n$ )  
   $y \leftarrow 1$   
  for  $i \leftarrow 0$  to  $n_b - 1$   
    if  $x_i = 1$  then  $y \leftarrow a \cdot y \bmod n$   
     $a \leftarrow a^2 \bmod n$   
  return  $y$ 
```

Theorem

If p is prime and $p \nmid a$, then

$$a^{p-1} \equiv 1 \pmod{p},$$

and hence

$$a^{-1} \equiv a^{p-2} \pmod{p}.$$

A second route to an inverse.

Example — inversion by Fermat

Invert 5 modulo 13 using Fermat and square-and-multiply.

Verify your answer.

Two ways to invert

Extended Euclid	Fermat
any modulus	needs a prime modulus
$O(\log^2)$ time	one exponentiation
branching	constant-time

Constant time matters for side-channel resistance. Both will be used.

Theorem

Let $m_1, \dots, m_k$ be pairwise coprime, $M = m_1 \cdots m_k$. Then

$$\psi : \mathbb{Z}_M \longrightarrow \mathbb{Z}_{m_1} \times \cdots \times \mathbb{Z}_{m_k},$$

$$\psi(x) = (x \bmod m_1, \dots, x \bmod m_k)$$

is a *ring isomorphism*: a bijection respecting both $+$ and $\times$ componentwise.

Why state it this way?

The usual treatment presents CRT as a *procedure* for solving a system of congruences.

That procedure is really the computation of ψ^{-1} .

Stating the theorem as a fact about the **structure of the ring** is what makes it reusable.

Computing the inverse map

Set $M_i = M/m_i$ and $y_i = M_i^{-1} \bmod m_i$, and define the **idempotents**

$$e_i = M_i y_i \bmod M.$$

Then

$$\psi^{-1}(a_1, \dots, a_k) = \left(\sum_i a_i e_i \right) \bmod M.$$

$$e_i \equiv 1 \pmod{m_i}, \quad e_i \equiv 0 \pmod{m_j} \ (j \neq i),$$

$$e_i^2 \equiv e_i, \quad \sum_i e_i \equiv 1 \pmod{M}.$$

Example — CRT

Take $m_1 = 5$, $m_2 = 13$, so $M = 65$.

Compute e_1 and e_2 and check their properties. Then find $\psi(23)$ and recover 23 from it.

One theorem, four instantiations

Instance	What it buys
$\mathbb{Z}_M \cong \prod \mathbb{Z}_{m_i}$	CRT secret sharing (L15)
$\mathbb{Z}_q[X]/(X^n+1) \cong \prod \mathbb{Z}_q[X]/(X^2-r_i)$	the NTT: $O(n \log n)$ multiplication
$\mathbb{Z}_Q \cong \prod \mathbb{Z}_{q_i}$	RNS: the BGV/CKKS modulus chain
$R_t \cong \prod \mathbb{F}_{t^d}$	batching: $n/2$ SIMD plaintext slots

We will meet all four.

Componentwise arithmetic is the point

Under ψ , multiplying two elements of $\mathbb{Z}_M$ becomes

k independent small multiplications.

Every practical FHE library represents its enormous ciphertext modulus this way.

Not everything is lattice-based

The zero-knowledge material, and the BGN scheme, rest on assumptions about **cyclic groups** instead.

Throughout: $\mathbb{G}$ is a cyclic group of *prime* order p with generator g .

Prime order is what makes these clean — every non-identity element generates the whole group.

Problem

Given g and $h = g^x$, find x .

Problem

Given g , g^a , g^b , compute g^{ab} .

Problem

Given g , g^a , g^b and a challenge z , decide whether

$z = g^{ab}$ or z is a uniformly random group element.

DDH hard $\implies$ CDH hard $\implies$ DLog hard

DDH is the strongest assumption; breaking DLog breaks all of them.

Definition

An efficiently computable map $e : \mathbb{G}_1 \times \mathbb{G}_2 \rightarrow \mathbb{G}_T$ between groups of prime order p such that

- i) **Bilinearity:** $e(u^a, v^b) = e(u, v)^{ab}$
- ii) **Non-degeneracy:** $e(g_1, g_2) \neq 1$

What a pairing buys — and costs

A pairing lets one multiply *in the exponent* exactly once.

- That one multiplication gives BGN its single homomorphic multiplication (L8).
- It also makes KZG polynomial commitments possible (ZKP material).
- But DDH becomes **easy** in $\mathbb{G}_1$ when a pairing into $\mathbb{G}_T$ exists.

A capability and a constraint at once.

Definition

$\nu : \mathbb{N} \rightarrow \mathbb{R}_{\geq 0}$ is **negligible** if for every constant $c > 0$ there is λ_0 with

$$\nu(\lambda) < \lambda^{-c} \quad \forall \lambda > \lambda_0.$$

Informally: smaller than any inverse polynomial.

Definition

For an adversary $\mathcal{A}$ distinguishing Exp_0 from Exp_1 ,

$$\text{Adv}(\mathcal{A}) = \left| \Pr[\mathcal{A}(\text{Exp}_0) = 1] - \Pr[\mathcal{A}(\text{Exp}_1) = 1] \right|.$$

A scheme is secure when $\text{Adv}(\mathcal{A})$ is negligible in λ for every efficient $\mathcal{A}$.

The security parameter

λ is the knob: $\lambda \in \{128, 192, 256\}$ in current standards.

Informal reading: an attack should require about 2^λ basic operations.

- ➊ Division gives a unique q, r once we insist $r \geq 0, n > 0$. Modular arithmetic keeps r and discards q .
- ➋ Extended Euclid gives Bézout; Bézout gives inverses. Works over any Euclidean domain — integers and polynomials alike.
- ➌ $\mathbb{Z}_n$ is a quotient ring. Reducing early is always legitimate. Additive inverses always exist; multiplicative ones exist exactly for elements coprime to n .

- ④ We use *centered* representatives, measure size with $\|\cdot\|_\infty$, and distinguish rounding from truncation. Every noise bound in the course is written in this language.
- ⑤ Fixed-point arithmetic is scaling by Δ plus rounding. Multiplication squares the scale and must be followed by a rescale — exactly what CKKS does homomorphically.

- 6 $\mathbb{F}_p^\times$ is cyclic of order $p - 1$; it contains an element of order d iff $d \mid p - 1$. This decides whether the NTT splits completely.
- 7 CRT is an isomorphism of rings, and we will instantiate it four times.
- 8 DLog / CDH / DDH and bilinear pairings support the non-lattice half of the course.

Lecture 2 — Polynomials over Finite Fields

Interpolation, identity testing, and the NTT.

Everything in this lecture, transplanted from $\mathbb{Z}$ to $\mathbb{F}_p[X]$.